

Wilkinsburg Land Bank

Executive Summary

Client

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Student Consulting Team

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Background

The Wilkinsburg Land Bank (WLB) is a nonprofit organization focused on transforming abandoned, tax-delinquent, and vacant properties into homes for new owners, with the goal of fostering racial and economic equity and enabling economic mobility for Wilkinsburg renters. Operating in a community with high vacancy rates and low homeownership, the organization works to acquire, stabilize, and transfer properties for long-term community benefit. However, limited staffing capacity and fragmented data systems have made property prioritization and decision-making challenging.

Project Description

Project Opportunity

Wilkinsburg faced a large number of vacant and abandoned properties, while the WLB had the capacity to address only a small subset each year. As a result, prioritization was the most critical decision point. Before this project, property selection relied on scattered datasets, manual lookups, and in-person inspections, making the process slow, inconsistent, and difficult to justify to stakeholders. The opportunity was to introduce a data-driven pre-screening approach to identify high-potential properties more efficiently and transparently.

Project Vision

The project aimed to design a system to make WLB's parcel decision making workflow more efficient. The solution focused on three components: a centralized dataset to consolidate property information, a transparent ranking and scoring system to evaluate redevelopment feasibility, and an interactive map interface that enables users to visualize and analyze parcels geographically within a single platform.

Project Outcomes

The project delivered a data-driven prioritization framework that improved efficiency, consistency, and transparency across people, process, and technology dimensions. From a people perspective, WLB's Executive Director Renee can now independently evaluate and prioritize parcels using a unified system, while the other board members benefit from clearer, data-backed justifications for decisions.

From a process perspective, the workflow shifted from manual, inspection-heavy evaluation to structured pre-screening, reducing parcel lookup time and enabling faster identification of high-priority properties. From a technology perspective, the solution integrates a centralized dataset, automated data collection through scraper workflows, and an interactive AppSheet interface with ranking and mapping features, creating a scalable and maintainable system aligned with existing tools. Additionally, the project established a foundation for integrating data into eProperty as a long-term system of record, improving data consistency, supporting structured reporting, and enabling future capabilities such as dashboards and public-facing tools.

Project Deliverables

The project deliverables include a unified parcel dataset that consolidates property, tax, and geographic data; a set of data scraper scripts that support ongoing updates; an eProperty dataset along with guidance for integrating and structuring data; an AppSheet interface with ranking, filtering, and mapping capabilities; and comprehensive documentation and training materials. Together, these deliverables provide the client with the tools and guidance needed to operate, maintain, and extend the system independently.

Recommendations

To sustain and expand the impact of the project, several recommendations are provided. These include establishing a regular data refresh process to maintain data accuracy, validating and refining the ranking model using historical outcomes, improving integration with eProperty for long-term scalability, and expanding spatial analysis to support neighborhood-level decision-making. Additionally, the report recommends exploring the use of AI tools, such as large language models, to automate property summaries and enhance communication with stakeholders. These steps will help the WLB continue to improve efficiency, strengthen decision-making, and increase its capacity to achieve its mission.

Student Consulting Team

Edison Dao is a third-year student majoring in Information Systems with a minor in Machine Learning. He served as the client relations lead and is interested in data-driven product and technical roles. He will graduate in 2027.

Luciana Requena is a third-year student majoring in Information Systems and Human-Computer Interaction, with a minor in Design. She served as the project manager and is interested in UI/UX design. She will be interning at Caterpillar Inc. this summer as a UI/UX designer.

Annie Zhang is a senior pursuing a dual degree in Information Systems and Linguistics, with minors in Computer Science and Photography. She served as the quality assurance lead and tech researcher. She will graduate in 2026 and is looking toward a career in software engineering.